Introduction to Linux Services Theory:

Understanding services in Linux: Systemd, init.d.Introduction to Apache and Nginx web servers.

Prcatical: Install Apache and start/stop services using systemctl.

Task: Verify the web server is running by accessing it in a browser.

TASKS

1.Install the Apache web server on your Linux machine using the package manager (apt or yum), and confirm the installation by running the command to check its version.

Install Apache Web Server

Step 1: Check Internet

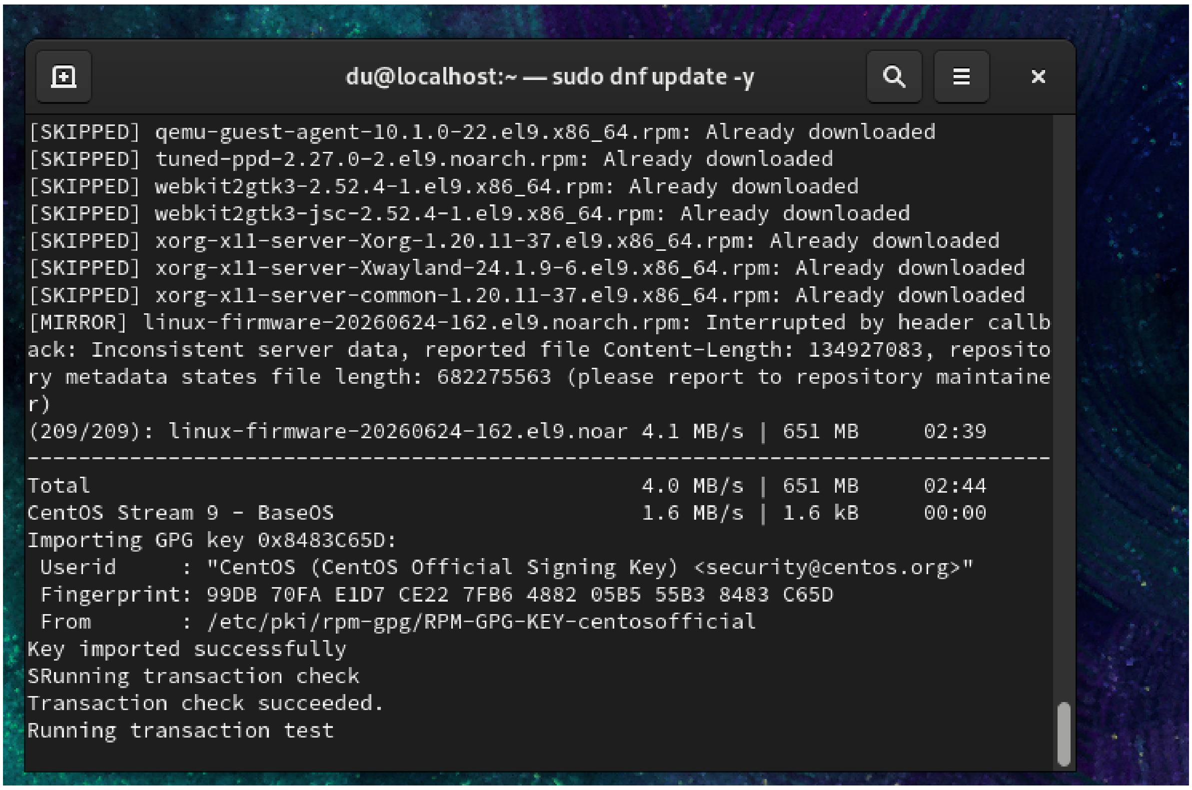
ping -c 4 8.8.8.8

```
[du@localhost ~]$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=82.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=93.7 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=29.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=223 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 29.859/107.335/222.935/70.979 ms
[du@localhost ~]$
```

Step 2: Update Packages

sudo dnf update -y

```
[du@localhost ~]$ sudo dnf update -y

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for du:
```

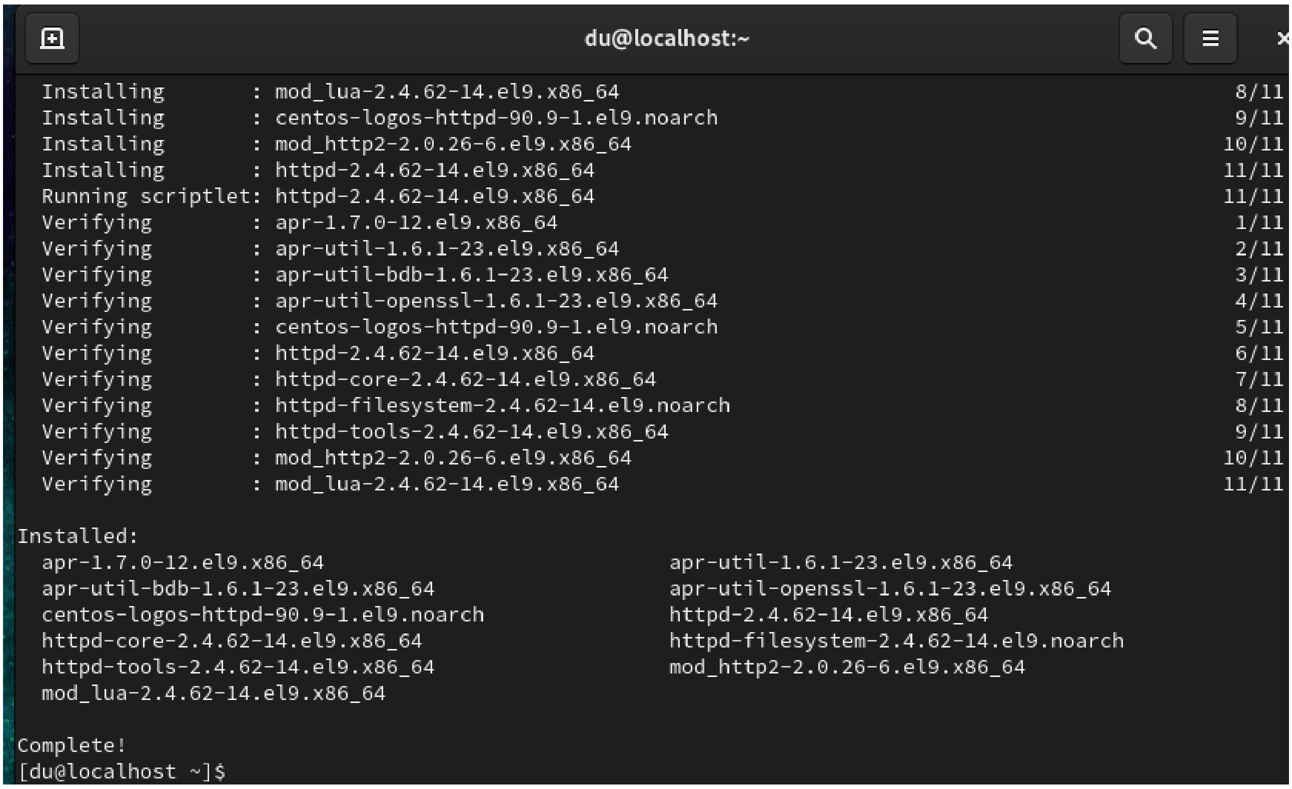


```
du@localhost:~ — sudo dnf update -y
[SKIPPED] qemu-guest-agent-10.1.0-22.el9.x86_64.rpm: Already downloaded
[SKIPPED] tuned-ppd-2.27.0-2.el9.noarch.rpm: Already downloaded
[SKIPPED] webkit2gtk3-2.52.4-1.el9.x86_64.rpm: Already downloaded
[SKIPPED] webkit2gtk3-jsc-2.52.4-1.el9.x86_64.rpm: Already downloaded
[SKIPPED] xorg-x11-server-Xorg-1.20.11-37.el9.x86_64.rpm: Already downloaded
[SKIPPED] xorg-x11-server-Xwayland-24.1.9-6.el9.x86_64.rpm: Already downloaded
[SKIPPED] xorg-x11-server-common-1.20.11-37.el9.x86_64.rpm: Already downloaded
[MIRROR] linux-firmware-20260624-162.el9.noarch.rpm: Interrupted by header callb
ack: Inconsistent server data, reported file Content-Length: 134927083, reposito
ry metadata states file length: 682275563 (please report to repository maintaine
r)
(209/209): linux-firmware-20260624-162.el9.noar 4.1 MB/s | 651 MB     02:39
--------------------------------------------------------------------------------
Total                                           4.0 MB/s | 651 MB     02:44
CentOS Stream 9 - BaseOS                        1.6 MB/s | 1.6 kB     00:00
Importing GPG key 0x8483C65D:
 Userid     : "CentOS (CentOS Official Signing Key) <security@centos.org>"
 Fingerprint: 99DB 70FA E1D7 CE22 7FB6 4882 05B5 55B3 8483 C65D
 From       : /etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
Key imported successfully
SRunning transaction check
Transaction check succeeded.
Running transaction test
```

```
[du@localhost ~]$ sudo dnf update -y
[sudo] password for du:
CentOS Stream 9 - BaseOS                                  1.9 kB/s | 5.4 kB     00:02
CentOS Stream 9 - AppStream                               3.3 kB/s | 5.5 kB     00:01
CentOS Stream 9 - Extras packages                         2.8 kB/s | 6.2 kB     00:02
Dependencies resolved.
Nothing to do.
Complete!
[du@localhost ~]$
```

Step 3: Install Apache

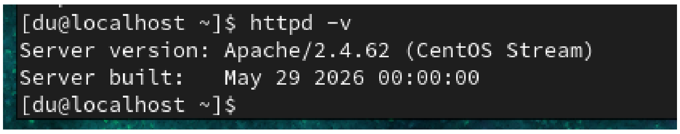
sudo dnf install httpd -y



```
du@localhost:~
  Installing        : mod_lua-2.4.62-14.el9.x86_64                    8/11
  Installing        : centos-logos-httpd-90.9-1.el9.noarch            9/11
  Installing        : mod_http2-2.0.26-6.el9.x86_64                  10/11
  Installing        : httpd-2.4.62-14.el9.x86_64                     11/11
  Running scriptlet: httpd-2.4.62-14.el9.x86_64                      11/11
  Verifying         : apr-1.7.0-12.el9.x86_64                          1/11
  Verifying         : apr-util-1.6.1-23.el9.x86_64                     2/11
  Verifying         : apr-util-bdb-1.6.1-23.el9.x86_64                 3/11
  Verifying         : apr-util-openssl-1.6.1-23.el9.x86_64             4/11
  Verifying         : centos-logos-httpd-90.9-1.el9.noarch             5/11
  Verifying         : httpd-2.4.62-14.el9.x86_64                       6/11
  Verifying         : httpd-core-2.4.62-14.el9.x86_64                  7/11
  Verifying         : httpd-filesystem-2.4.62-14.el9.noarch            8/11
  Verifying         : httpd-tools-2.4.62-14.el9.x86_64                 9/11
  Verifying         : mod_http2-2.0.26-6.el9.x86_64                   10/11
  Verifying         : mod_lua-2.4.62-14.el9.x86_64                    11/11

Installed:
  apr-1.7.0-12.el9.x86_64                    apr-util-1.6.1-23.el9.x86_64
  apr-util-bdb-1.6.1-23.el9.x86_64           apr-util-openssl-1.6.1-23.el9.x86_64
  centos-logos-httpd-90.9-1.el9.noarch       httpd-2.4.62-14.el9.x86_64
  httpd-core-2.4.62-14.el9.x86_64            httpd-filesystem-2.4.62-14.el9.noarch
  httpd-tools-2.4.62-14.el9.x86_64           mod_http2-2.0.26-6.el9.x86_64
  mod_lua-2.4.62-14.el9.x86_64

Complete!
[du@localhost ~]$
```

Step 4: Verify Apache Version

httpd -v



```
[du@localhost ~]$ httpd -v
Server version: Apache/2.4.62 (CentOS Stream)
Server built:   May 29 2026 00:00:00
[du@localhost ~]$
```

2.Start the Apache service using systemctl, then use systemctl status to verify it is running.

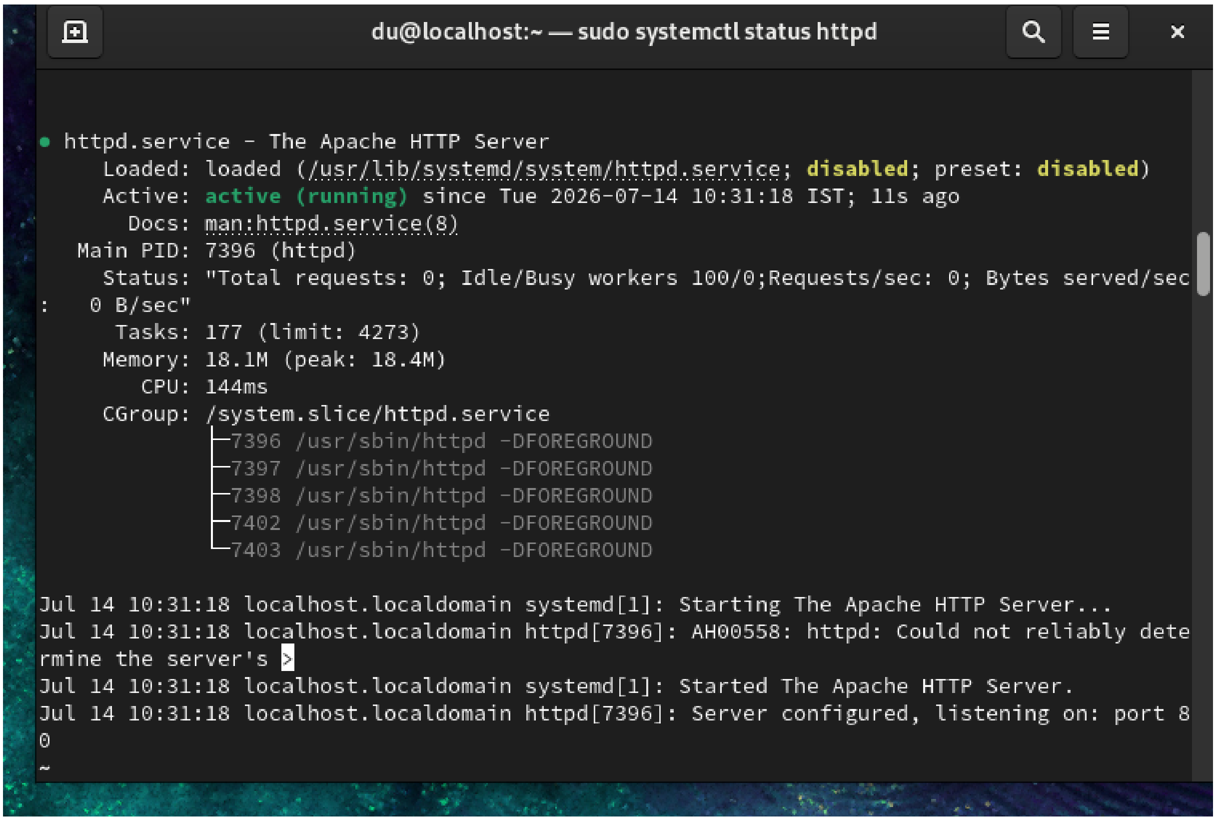
Start Apache Service

Start the service:

sudo systemctl start httpd

Check status:
sudo systemctl status httpd

Press **Q** to exit.



**3.Stop and restart the Apache service using systemctl, and observe the difference between stop, start, and restart commands.

Hint: Use 'systemctl stop apache2', 'systemctl start apache2', and 'systemctl restart apache2' (or 'httpd' on some systems).**

Stop and Restart Apache

Stop

sudo systemctl stop httpd

Verify:
sudo systemctl status httpd

```
[du@localhost ~]$ sudo systemctl stop httpd
[du@localhost ~]$ sudo systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:httpd.service(8)

Jul 14 10:31:18 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Jul 14 10:31:18 localhost.localdomain httpd[7396]: AH00558: httpd: Could not reliably det>
Jul 14 10:31:18 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
Jul 14 10:31:18 localhost.localdomain httpd[7396]: Server configured, listening on: port >
Jul 14 10:34:38 localhost.localdomain systemd[1]: Stopping The Apache HTTP Server...
Jul 14 10:34:40 localhost.localdomain systemd[1]: httpd.service: Deactivated successfully.
Jul 14 10:34:40 localhost.localdomain systemd[1]: Stopped The Apache HTTP Server.
lines 1-12/12 (END)
```

Start Again

sudo systemctl start httpd

Verify:

sudo systemctl status httpd

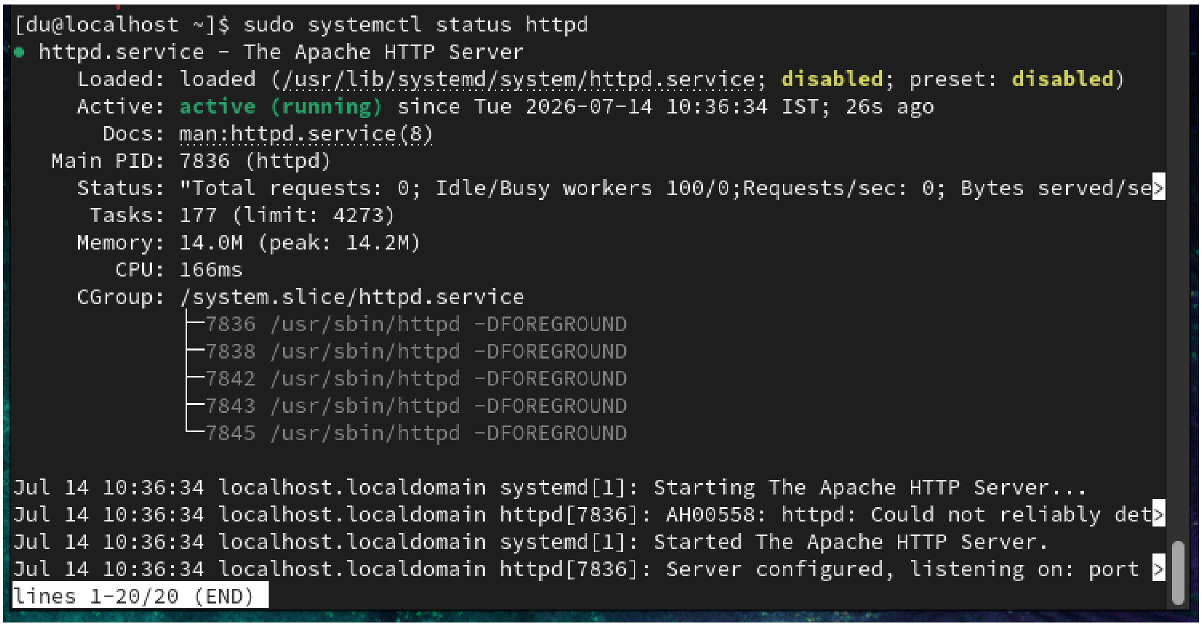
```
[du@localhost ~]$ sudo systemctl start httpd
[du@localhost ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Tue 2026-07-14 10:35:49 IST; 10s ago
       Docs: man:httpd.service(8)
   Main PID: 7635 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/se>
      Tasks: 177 (limit: 4273)
     Memory: 14.0M (peak: 14.3M)
        CPU: 115ms
     CGroup: /system.slice/httpd.service
             ├─7635 /usr/sbin/httpd -DFOREGROUND
             ├─7636 /usr/sbin/httpd -DFOREGROUND
             ├─7640 /usr/sbin/httpd -DFOREGROUND
             ├─7641 /usr/sbin/httpd -DFOREGROUND
             └─7643 /usr/sbin/httpd -DFOREGROUND

Jul 14 10:35:49 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Jul 14 10:35:49 localhost.localdomain httpd[7635]: AH00558: httpd: Could not reliably det>
Jul 14 10:35:49 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
Jul 14 10:35:49 localhost.localdomain httpd[7635]: Server configured, listening on: port >
lines 1-20/20 (END)
```

Restart

sudo systemctl restart httpd

Verify:

sudo systemctl status httpd



```
[du@localhost ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Tue 2026-07-14 10:36:34 IST; 26s ago
       Docs: man:httpd.service(8)
   Main PID: 7836 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/se>
      Tasks: 177 (limit: 4273)
     Memory: 14.0M (peak: 14.2M)
        CPU: 166ms
     CGroup: /system.slice/httpd.service
             ├─7836 /usr/sbin/httpd -DFOREGROUND
             ├─7838 /usr/sbin/httpd -DFOREGROUND
             ├─7842 /usr/sbin/httpd -DFOREGROUND
             ├─7843 /usr/sbin/httpd -DFOREGROUND
             └─7845 /usr/sbin/httpd -DFOREGROUND

Jul 14 10:36:34 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Jul 14 10:36:34 localhost.localdomain httpd[7836]: AH00558: httpd: Could not reliably det>
Jul 14 10:36:34 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
Jul 14 10:36:34 localhost.localdomain httpd[7836]: Server configured, listening on: port >
lines 1-20/20 (END)
```

4.Open your browser and access http://localhost or your Linux machine's IP address to verify that the Apache default web page is displayed.

Verify in Browser

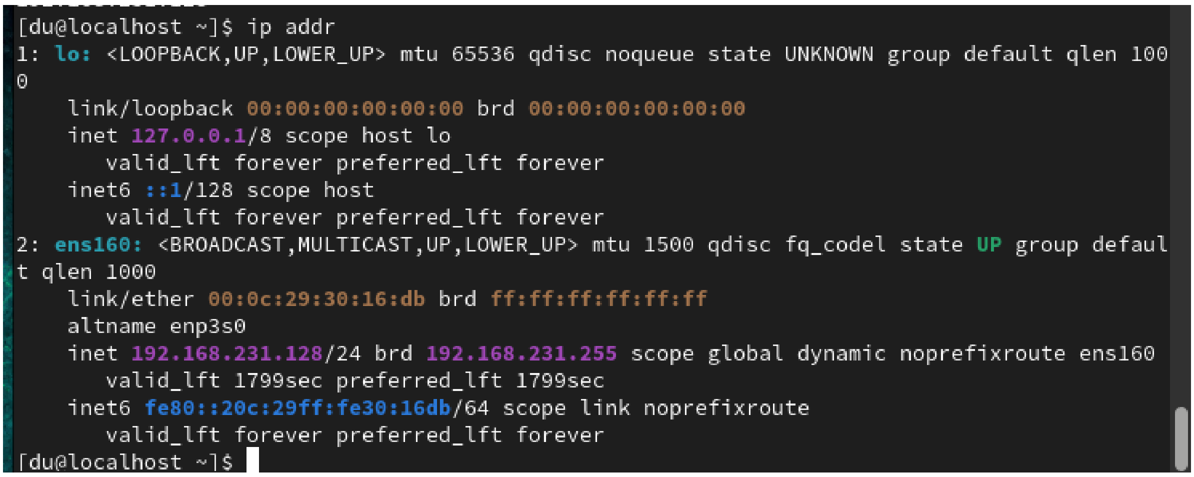
Check your IP:

hostname -I

```
[du@localhost ~]$ hostname -I
192.168.231.128
```

Or

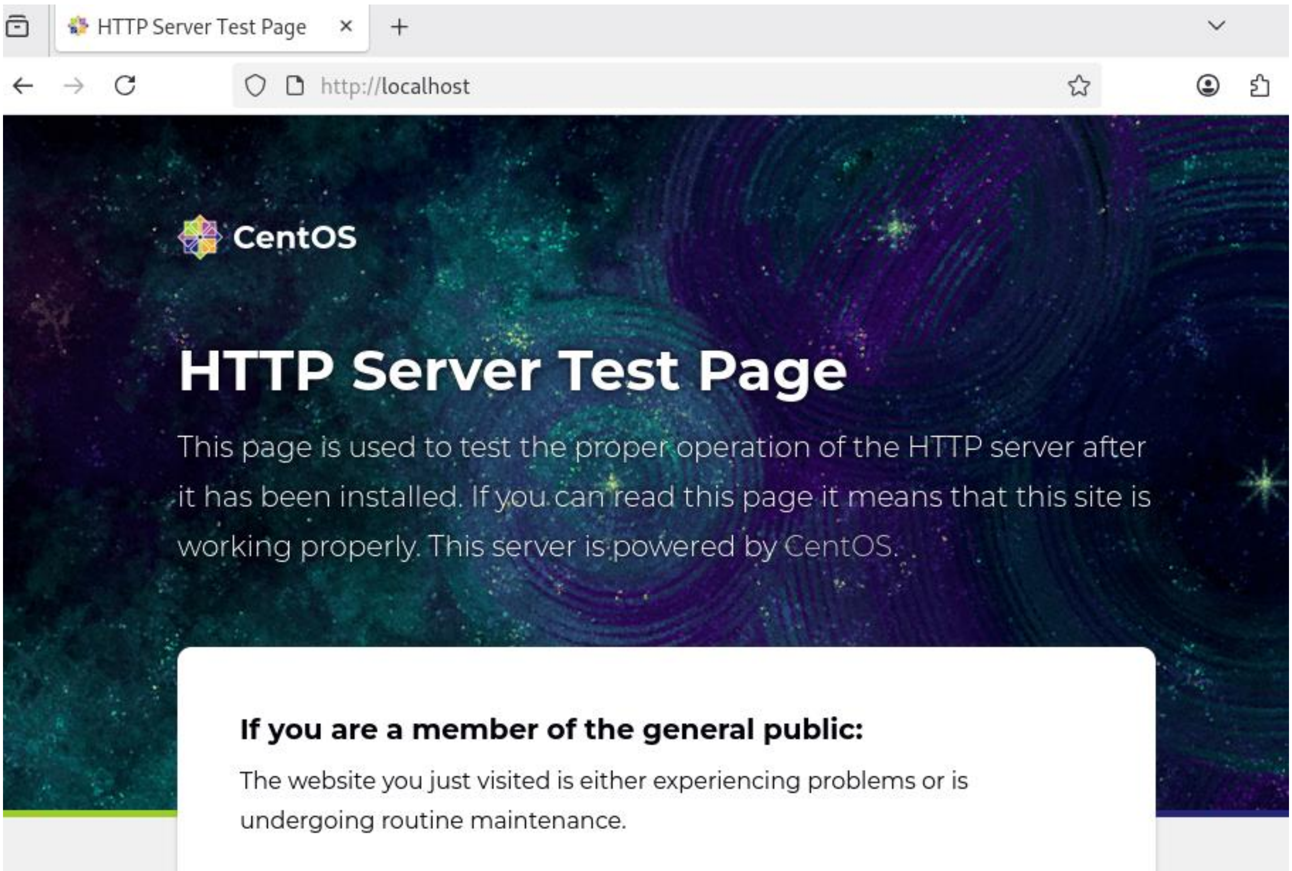
ip addr



```
[du@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 100
0
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group defaul
t qlen 1000
    link/ether 00:0c:29:30:16:db brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.231.128/24 brd 192.168.231.255 scope global dynamic noprefixroute ens160
       valid_lft 1799sec preferred_lft 1799sec
    inet6 fe80::20c:29ff:fe30:16db/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[du@localhost ~]$
```

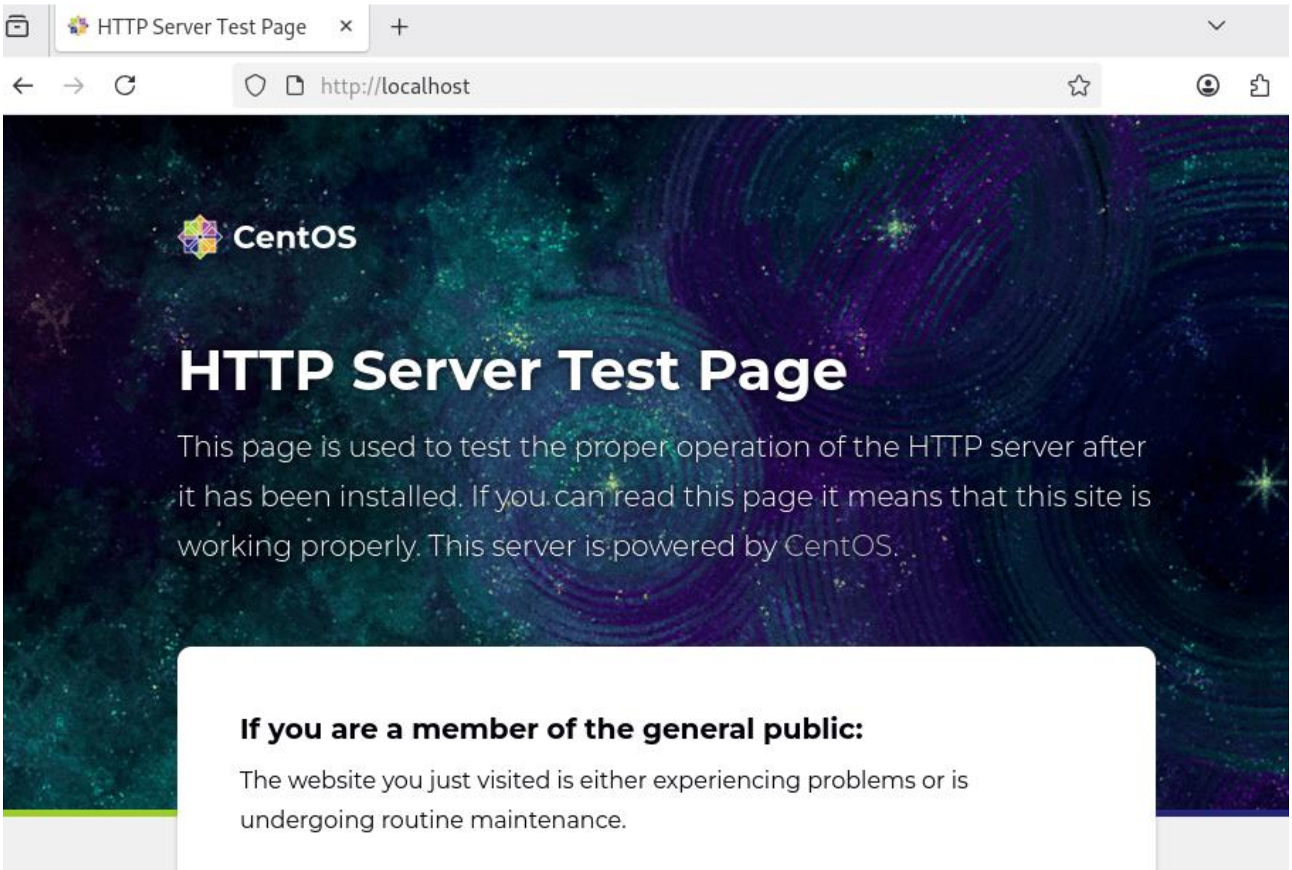
Open browser:

http://localhost



Or

http://192.168.75.128



If Firewall is Running

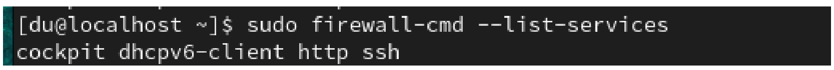
Open HTTP service:

sudo firewall-cmd --permanent –add-service=http

sudo firewall-cmd –reload

Verify:

sudo firewall-cmd --list-services



5.Write a short comparison (3-4 lines) between Apache and Nginx web servers, mentioning one key use-case for each based on your research.

Apache

Apache is an open-source web server that is easy to configure and supports many modules. It is suitable for dynamic websites and is widely used in web hosting.

Nginx

Nginx is a high-performance web server that handles many concurrent connections efficiently. It is commonly used for high-traffic websites and as a reverse proxy.